MOVIE TICKET BOOKING SYSTEM

A PROJECT REPORT

Submitted by

SUJITKUMAAR G S

in partial fulfilment for the award of the course

CGB1221 – DATABASE MANAGEMENT SYSTEMS

IN

DEPARTMENT OF

COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING**)**

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
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CHENNAI 600 025

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PROJECT FINAL DOCUMENT

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Under the Guidance of

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BONAFIDE CERTIFICATE

Certified that this project report titled **"MOVIE TICKET BOOKING SYSTEM" is** the bonafide work of **SUJITKUMAAR G S (8115U24AM054)** who carried out the work under my supervision.

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DECLARATION BY THE CANDIDATE

I declare that to the best of my knowledge the work reported here in has been composed solely by myself and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva-Voice held at K. Ramakrishnan College of Engineering on __________

SIGNATURE OF THE CANDIDATE

ACKNOWLEDGEMENT

I thank the almighty GOD, without whom it would not have been possible for me to complete my project.

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DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

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To become a renowned hub for AIML technologies to producing highly talented globally recognizable technocrats to meet industrial needs and societal expectation.

MISSION

Mission of the Department

M1 To impart advanced education in AI and Machine Learning, built upon a foundation in Computer Science and Engineering.

M2 To foster Experiential learning equips students with engineering skills to tackle real-world problems.

M3 To promote collaborative innovation in AI, machine learning, and related research and development with industries.

M4 To provide an enjoyable environment for pursuing excellence while upholding strong personal and professional values and ethics.

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PEO1 Excel in technical abilities to build intelligent systems in the fields of AI & ML in order to find new opportunities.

PEO2 Embrace new technology to solve real-world problems, whether alone or as a team, while prioritizing ethics and societal benefits.

PEO3 Accept lifelong learning to expand future opportunities in research and product development.

PROGRAM SPECIFIC OUTCOMES (PSO's)

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PSO2 Ability to conduct research, contributing to machine learning advancements and innovations that tackle emerging societal challenges.

PROGRAM OUTCOMES (POs)

Engineering students will be able to:

PO1 Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems. **PO2 Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3 Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4 Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6 The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO8 Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings

PO9 Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10 Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11 Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

The **Movie Ticket Booking System** is a web-based application designed to allow users to browse movies, view show timings, select seats, and book tickets efficiently. The platform provides a clean and responsive interface where users can explore currently running movies, check theater availability, and make bookings in real time. It uses HTML, CSS, and JavaScript for the frontend, with a DBMS such as MySQL for backend data management. Each movie, theater, show schedule, user profile, and booking transaction is stored in a normalized relational database to ensure data integrity and reduce redundancy. User authentication is secured with encrypted passwords to prevent unauthorized access. The admin module enables management of movies, show timings, and booking records while also monitoring system activities. CRUD operations are implemented throughout the system to handle dynamic data such as ticket booking, cancellation, and updates. The system also incorporates basic performance optimization and scalability considerations to handle multiple users simultaneously. It is lightweight and suitable for real-time applications in cinemas or online booking platforms. The project highlights the importance of DBMS in managing transactional systems efficiently. It provides a user-friendly and reliable environment for seamless movie ticket booking and management.

ABSTRACT WITH POs AND PSOs MAPPING

ABSTRACT	POs MAPPED	PSOs MAPPED
The Movie Ticket Booking System is a web-based application designed to allow users to browse movies, select show timings, book tickets, and manage their reservations conveniently. The system supports user registration, secure login, movie browsing with details such as show time and theater information, seat selection, and ticket booking functionality. It also includes an admin panel for managing movies, theaters, show schedules, and booking records. The platform is built using a **Relational Database Management System (RDBMS)** to efficiently handle user data, movie details, show timings, seat availability, and booking transactions. It aims to provide a simple and efficient online ticket reservation experience through an intuitive user interface. Key features include CRUD operations on movies and bookings, user role management for customers and administrators, responsive design for multi-device support, real-time seat availability updates, and secure session handling to ensure safe and reliable transactions.	**PO1-3** **PO2-3** **PO3-3** **PO4-3** **PO5-3** **PO6-3** **PO7-3** **PO8-3** **PO9-3** **PO10-3** **PO113** **PO12-3**	**PSO1 – 3** **PSO2 – 3**

Note: 1- Low, 2-Medium, 3- High

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LIST OF ABBREVIATIONS

S. NO.	ACRONYM	ABBREVIATION
1	AJAX	Asynchronous JavaScript and XML
2	API	Application Programming Interface
3	CRUD	Create, Read, Update, Delete
4	CSS	Cascading Style Sheets
5	DBMS	Database Management System
6	HTML	Hyper Text Markup Language
7	HTTP	HyperText Transfer Protocol
8	JS	JavaScript
9	SQL	Structured Query Language

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LIST OF FIGURES

CHAPTER 1
INTRODUCTION

1.1 OBJECTIVE

The primary objective of the Movie Ticket Booking System project is to design and develop a user-friendly web application that allows users to conveniently browse movies, check show timings, select seats, and book tickets online. In today's fast-paced digital world, online ticket booking systems have become essential for providing quick and hassle-free services to customers, eliminating long queues at theaters.

This system aims to provide a structured and efficient platform for managing movie schedules, theater details, and booking transactions. It is developed with the goal of implementing all essential CRUD (Create, Read, Update, Delete) operations for movies, users, and bookings. Users can register and log in securely, explore available movies, view show timings, and book or cancel tickets easily.

The system also enhances user experience by providing real-time seat availability and booking confirmation. Additionally, it includes an admin module that allows administrators to manage movies, theaters, show schedules, and user bookings. Admins can update movie details, add new shows, and monitor system activities to ensure smooth functioning.

Security is an important aspect of the system. User authentication is implemented with encrypted passwords to prevent unauthorized access. The system is also designed with scalability in mind, allowing future enhancements such as online payment integration, seat recommendations, and mobile application support.

1.2 SCOPE OF THE PROJECT

The Movie Ticket Booking System is designed as a dynamic and efficient web-based application that simplifies the process of booking movie tickets. The scope of the project covers a wide range of functionalities suitable for users such as customers, theater staff, and administrators.

At its core, the system allows users to register, log in, browse movies, check show timings, select seats, and book tickets through an intuitive interface. Users can also view their booking history and cancel tickets if required. The system ensures a smooth and user-friendly experience for all users.

The system includes administrative features that allow admins to manage the entire platform. Admins can add, update, or delete movies, manage theater details, schedule shows, and monitor bookings. This ensures proper control and efficient management of the system.

The scope of the project includes the following modules:

- User Management Module: Handles user registration, login, authentication, and session management.
- Movie Management Module: Allows admin to add, update, and delete movie details.
- Show Management Module: Manages show timings and theater schedules.
- Booking Module: Enables users to select seats, book tickets, and cancel bookings.
- Admin Control Module: Provides full control over movies, users, and booking data.

From a technical perspective, the system integrates frontend technologies like HTML, CSS, and JavaScript with backend technologies (PHP/Python/Node.js) and a relational DBMS such as MySQL. It ensures efficient data handling through normalization and maintains data integrity using constraints and relationships.

This project can be extended for real-world applications such as cinema booking platforms, event ticketing systems, or entertainment portals. Overall, the scope includes both current functionalities and future enhancements, making it a valuable academic and practical project.

1.3 OVERVIEW OF THE SYSTEM

The Movie Ticket Booking System is a full-stack web application designed to provide users with a seamless and efficient way to book movie tickets online. The system operates on a client-server architecture, where users interact with the application through a web browser, and the backend server processes requests, performs necessary operations, and communicates with the database to store and retrieve information. This system is developed using a combination of frontend technologies, server-side scripting, and a relational database management system (DBMS), ensuring smooth performance and reliable data handling.

The platform supports two main types of users: customers and administrators. Customers can register and log in securely to access the system, browse available movies, view detailed information such as genre, duration, and ratings, and check show timings across different theaters. They can select seats based on real-time availability and proceed with ticket booking. The system also allows users to manage their bookings by viewing booking history and canceling tickets when required, providing flexibility and convenience.

On the administrative side, the system includes a dedicated admin dashboard that enables efficient management of the entire platform. Administrators have the authority to add, update, or delete movie details, manage theater information, schedule show timings, and monitor all booking transactions. This ensures that the system remains updated and operates smoothly without inconsistencies.

The user interface of the system is designed to be simple, intuitive, and responsive, allowing users to access the platform from various devices such as desktops, tablets, and smartphones. The backend of the system handles core functionalities including business logic, user authentication, booking validation, and database communication. It ensures that all operations such as ticket booking and cancellation are processed accurately and efficiently.

The database layer plays a crucial role in storing structured data such as user details, movie information, theater data, show schedules, and booking records. Relationships between different entities are maintained using foreign keys, and normalization techniques are applied to reduce redundancy and improve data consistency. Additionally, the system incorporates essential security features such as password encryption, session management, input validation, and protection against SQL injection to safeguard user data and maintain system integrity.

Overall, the Movie Ticket Booking System provides a reliable, secure, and user-friendly platform for online ticket booking. It enhances the user experience by offering real-time services and efficient data management while ensuring scalability and performance, making it suitable for real-world applications in the entertainment industry.

1.4 IMPORTANCE OF DBMS IN MOVIE TICKET BOOKING SYSTEM

A Database Management System (DBMS) plays a vital role in the effective functioning of the Movie Ticket Booking System, as it is responsible for managing and organizing large volumes of data generated by users, movies, theaters, show schedules, and booking transactions. In such a dynamic system where multiple users interact simultaneously, the DBMS ensures that data is stored in a structured, consistent, and secure manner. It provides a reliable mechanism for performing operations such as storing new booking records, retrieving movie details, updating seat availability.

In this system, the DBMS maintains separate tables for different entities such as users, movies, theaters, shows, and bookings, while establishing relationships between them using primary and foreign keys. This helps in accurately representing real-world relationships, such as one user booking multiple tickets or one movie having multiple show timings. By enforcing data integrity constraints like NOT NULL, UNIQUE, and referential integrity, the DBMS prevents invalid or duplicate data from being stored, ensuring accuracy and consistency across the system.

Security is another critical aspect where the DBMS plays an important role. Sensitive user data, especially login credentials, is protected through encryption and secure storage mechanisms. The DBMS also supports access control, ensuring that only authorized users, such as administrators, can modify or delete critical data. Additionally, it provides backup and recovery features that help restore data in case of system failures, thereby preventing data loss.

The DBMS also enhances system performance and scalability. With the help of indexing and optimized SQL queries, data retrieval becomes faster even when the number of users and bookings increases significantly. It allows the system to handle multiple transactions simultaneously without conflicts, ensuring a smooth booking experience for users. Features like real-time seat availability, booking history tracking, and report generation are made possible through efficient database operations.

Furthermore, normalization techniques are applied in the database design to eliminate redundancy and improve efficiency. This ensures that data is stored only once and updated consistently across the system. Overall, the DBMS acts as the backbone of the Movie Ticket Booking System, enabling it to operate smoothly, securely, and efficiently. Without a robust DBMS, managing such a complex and data-intensive application would be extremely difficult, making it an essential component of the system.

1.5 TECHNOLOGIES USED

The Movie Ticket Booking System is developed using a combination of front-end, back-end, and database technologies. Each layer of the application—presentation, logic, and data storage—is built using tools and frameworks that are reliable, secure, and scalable. These technologies work together to deliver a responsive, interactive, and fully functional web-based ticket booking system.

The main technologies used in this project are categorized as follows:

1. Front-End Technologies (Client Side)

HTML

HTML is used to design and structure the web pages of the system. It defines elements such as movie listings, booking forms, login pages, seat selection layouts, and navigation menus.

CSS

CSS is used to enhance the visual appearance of the application. It provides styling features such as colors, fonts, layouts, spacing, and responsive design to ensure a smooth user experience across different devices.

JavaScript

JavaScript adds interactivity and dynamic functionality to the system. It is used for real-time form validation, updating seat availability, handling user actions like booking and cancellation, and improving overall user experience.

Frontend Frameworks (Optional)

Frameworks like Bootstrap are used for responsive design and UI components. ReactJS can also be used to create reusable components and manage application state efficiently.

2. Back-End Technologies (Server Side)

PHP / Python (Flask/Django) / Node.js

These server-side technologies handle the core logic of the application. They process user requests, manage authentication, handle booking operations, and communicate with the database.

Authentication Tools

User authentication is secured using password hashing techniques such as bcrypt. Session management is implemented using cookies or tokens (JWT) to maintain user sessions securely.

Routing & Architecture

The system follows structured architecture patterns like Model-View-Controller (MVC), which separates business logic, user interface, and data handling, making the application easier to maintain and scale.

3. Database Management System

MySQL / PostgreSQL

A relational database is used to store structured data such as user details, movies, theaters, show timings, seat availability, and booking records. These DBMS tools are reliable and support ACID properties.

SQL Queries

Structured Query Language (SQL) is used to perform operations such as INSERT, SELECT, UPDATE, and DELETE. Advanced queries help in managing bookings, retrieving movie schedules, and generating reports.

4. Development Tools and Libraries

XAMPP / WAMP / Local Server

Used to run the application locally with a web server and database support.

VS Code

A modern code editor used for writing, editing, and debugging code efficiently.

GitHub

Used for version control, project backup, and collaboration.

Email Integration

SMTP services can be used to send booking confirmations, notifications, and password reset links.

5. Optional Enhancements

- Online payment integration (UPI, debit/credit cards).
- SMS or email notifications for booking confirmation.
- Deployment using cloud platforms like AWS, Heroku.
- Search and filter options for movies and show timings.
- Real-time seat availability updates using AJAX or APIs.

CHAPTER 2

LITERATURE SURVEY

This section reviews the concepts, methodologies, and system design approaches used in the development of the Movie Ticket Booking System. The system is developed following a structured and iterative methodology based on the principles of the Software Development Life Cycle (SDLC). The development process includes requirement analysis, system design, database modeling, implementation, and testing. The main objective is to build a reliable, efficient, and user-friendly application that simplifies the process of booking movie tickets online. This chapter explains the planning, analysis, and design aspects that form the foundation of the system.

2.1 REQUIREMENT ANALYSIS

The development of the Movie Ticket Booking System begins with identifying both functional and non-functional requirements, which define the behavior of the system and its operational constraints. The functional requirements of the system include user registration and secure login, browsing available movies, viewing show timings and theater details, selecting seats, booking tickets, and managing bookings such as viewing history or canceling tickets. The system also includes administrative functionalities such as adding, updating, or deleting movie details, managing show schedules, monitoring bookings, and controlling user access. In addition to these, the system provides responsive navigation and real-time updates of seat availability to enhance user experience. The non-functional requirements focus on the quality attributes of the system. The system is designed to be user-friendly, with a simple and intuitive interface that allows users to easily navigate and perform booking operations. Performance is an important factor, ensuring fast data retrieval and minimal response time during booking transactions.

2.2 ENTITY-RELATIONSHIP (ER) DIAGRAM

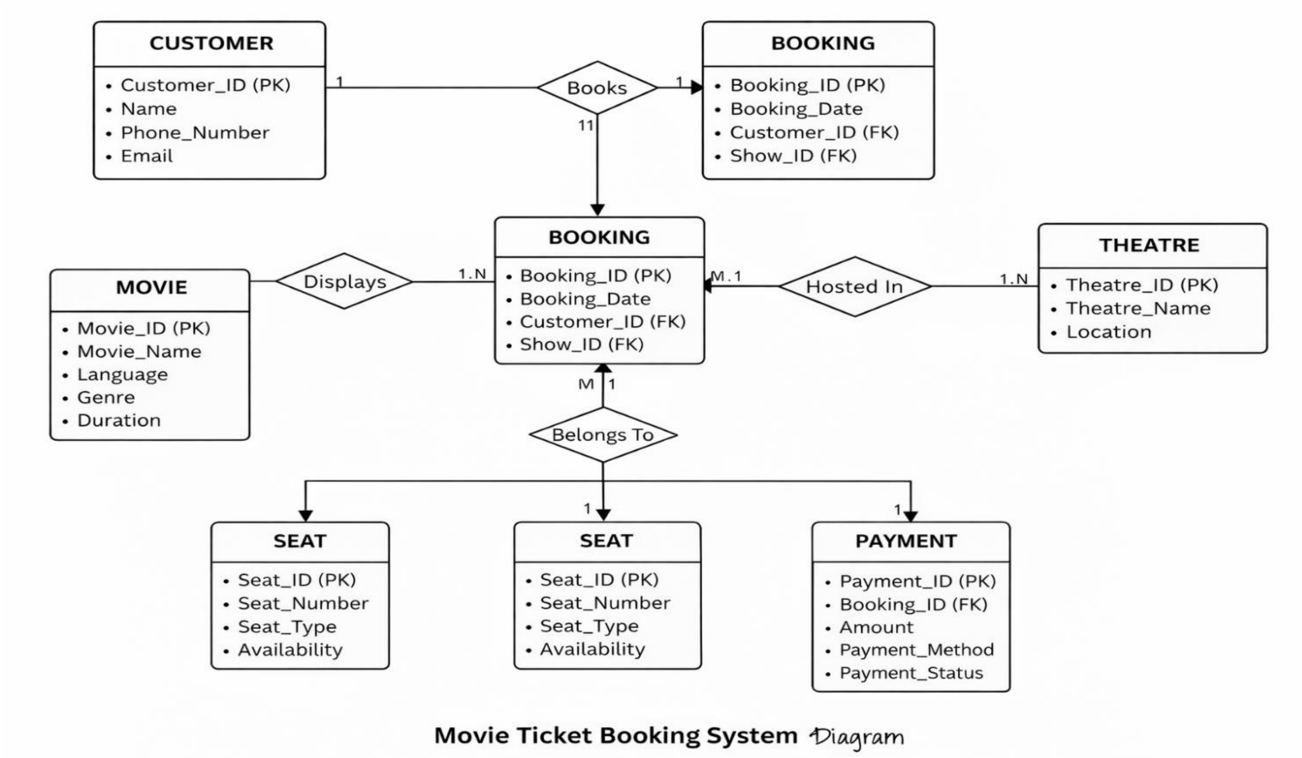

Figure 2.2.1 Entity Relationship

The Entity-Relationship (ER) Diagram of the Movie Ticket Booking System represents the structure of the database by showing the main entities and their relationships. The key entities include Customer, Movie, Booking, Theatre, Seat, and Payment. The Customer entity stores user details, while the Movie entity contains information about movies. The Booking entity acts as the central component, linking customers to movies and shows, and includes details such as booking date and references to customer and show IDs. The Theatre entity stores theater information, and the Seat entity manages seat details and availability. The Payment entity handles transaction-related information for each booking. The relationships indicate that one customer can make multiple bookings, a movie can have multiple bookings, and each booking is associated with a specific theatre, seat, and payment. This ER diagram helps in organizing data efficiently and ensures proper relationships between different components of the system.

2.3 RELATIONAL SCHEMA

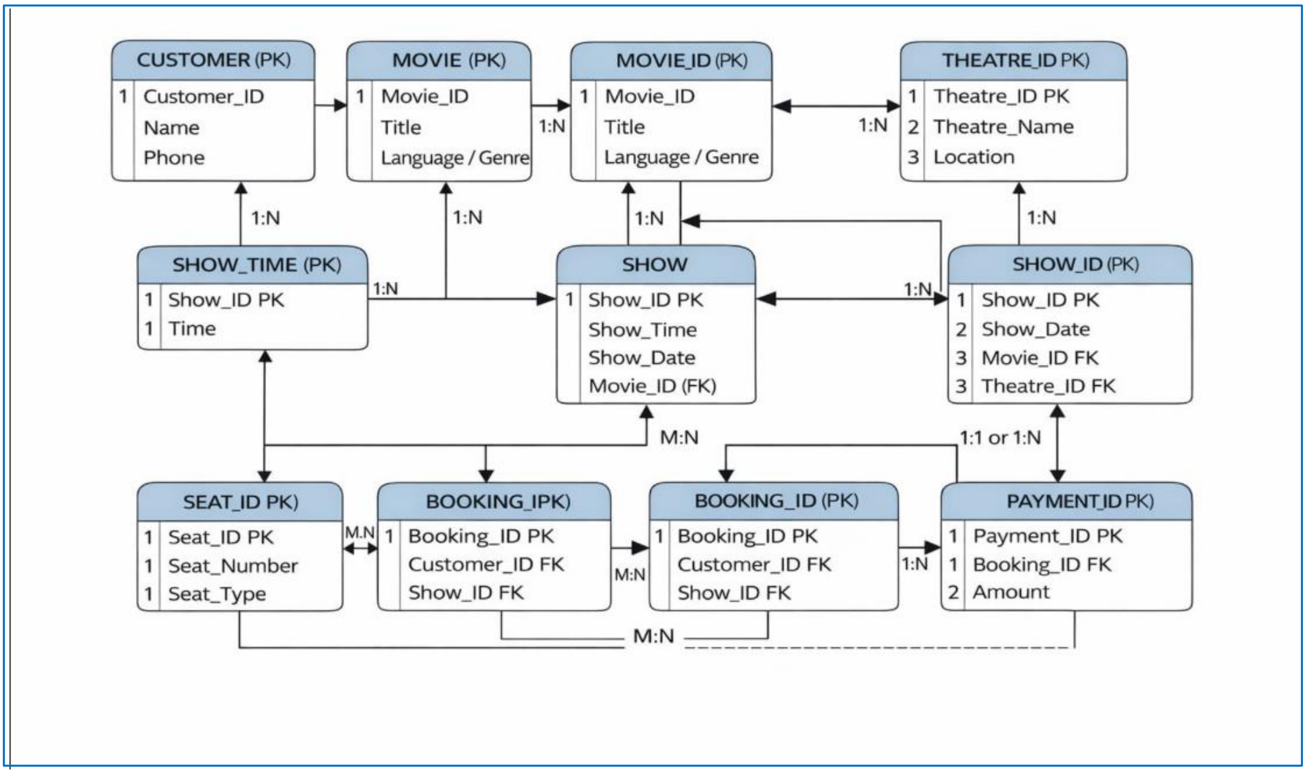

Figure 2.3.1 Relational Schema

The relational schema of the Movie Ticket Booking System defines how the data is organized into tables based on the ER diagram and how these tables are interconnected using keys. Each entity is represented as a table with attributes, where primary keys uniquely identify each record and foreign keys establish relationships between tables. The main tables include Customer(Customer_ID, Name, Phone_Number, Email), Movie(Movie_ID, Movie_Name, Language, Genre, Duration), Theatre(Theatre_ID, Theatre_Name, Location), Booking(Booking_ID, Booking_Date, Customer_ID, Show_ID), Seat(Seat_ID, Seat_Number, Seat_Type, Availability), and Payment(Payment_ID, Booking_ID, Amount, Payment_Method, Payment_Status). In this schema, Customer_ID in the Booking table acts as a foreign key linking customers to their bookings, while Booking_ID in the Payment table connects payments to bookings.

2.4 NORMALIZATION PROCESS

Normalization is the process of organizing data in the Movie Ticket Booking System to reduce redundancy and ensure efficient data management. The database is designed by applying the first three normal forms. In the First Normal Form (1NF), all attributes are made atomic, meaning each field contains only a single value and there are no repeating groups. In the Second Normal Form (2NF), all non-key attributes are fully dependent on the primary key, eliminating partial dependencies and ensuring that each table represents a single entity. In the Third Normal Form (3NF), transitive dependencies are removed so that non-key attributes depend only on the primary key, improving data consistency. For example, customer details, movie details, booking information, and payment data are stored in separate tables and connected through foreign keys, avoiding duplication of data. This normalization process results in a well-structured, efficient, and scalable database that improves performance and simplifies maintenance.

2.5 SYSTEM ARCHITECTURE OVERVIEW

The Movie Ticket Booking System follows a three-tier architecture consisting of the Frontend, Backend, and Database layers, each responsible for specific functionalities to ensure smooth system operation. These layers work together to handle user interactions, process business logic, and manage data efficiently.

The **Frontend Layer (Presentation Layer)** is developed using HTML, CSS, and JavaScript, and it serves as the interface between the user and the system. It allows customers and administrators to interact with the application through features such as movie selection, seat booking, login, and navigation. This layer is designed to be responsive and user-friendly, ensuring accessibility across different devices. It captures user inputs and sends requests to the backend for further processing.

The **Backend Layer (Application Layer)** is implemented using technologies such as Python, Java, or Node.js, and it handles the core logic of the system. This layer processes user requests received from the frontend, performs operations like booking validation, user authentication, and business logic execution, and sends responses back to the frontend. It also manages functionalities such as booking logic, user authentication, and notifications. The backend acts as a bridge between the frontend and the database, ensuring secure and efficient data processing.

The **Database Layer (Data Layer)** uses relational database systems such as MySQL or PostgreSQL to store and manage all persistent data. It includes data related to users, movies, show timings, seat availability, and bookings. This layer ensures data integrity, consistency, and efficient retrieval using SQL queries. It supports data storage, retrieval, and updates required by the backend layer.

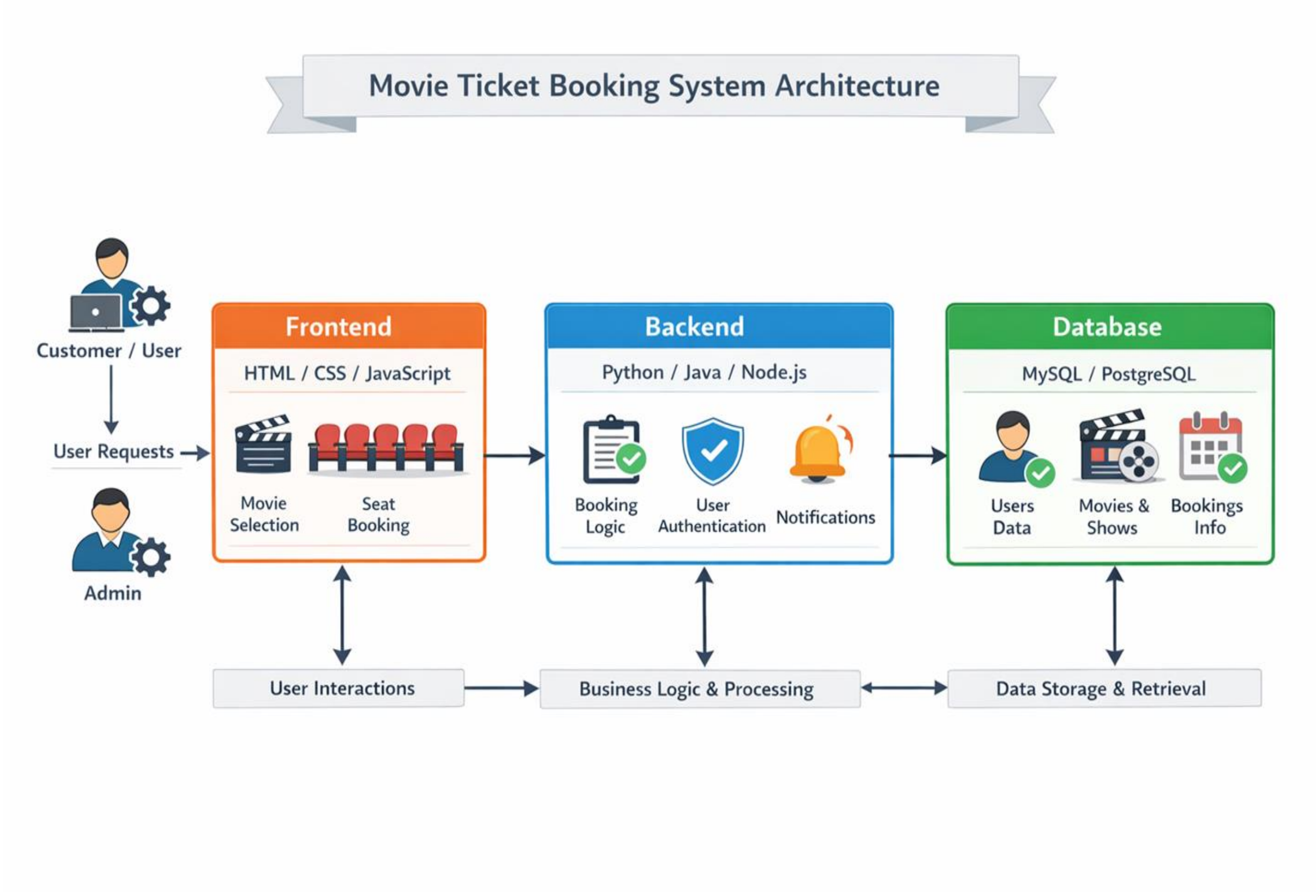

Figure 2.5.1 System Architecture

CHAPTER 3

PROJECT METHODOLOGY

The **Movie Ticket Booking System** is structured as a modular web application where each module is responsible for a specific functionality. This modular approach ensures separation of concerns, easy maintenance, and scalability. Each module interacts with other components of the system to provide a seamless and efficient ticket booking experience. This chapter describes the major modules of the system, their features, and their functional flow.

3.1 USER MANAGEMENT MODULE

The User Management Module handles all functionalities related to user accounts, including registration, login, logout, session handling, and role management. It ensures that only authenticated users can access booking features and perform operations securely.

Key Features:

• **User Registration:** Allows new users to sign up by providing details such as name, email, and password. Passwords are securely encrypted before storing in the database.

• **Login/Logout:** Validates user credentials and manages sessions for secure access.

• **Role Management:** Differentiates between customers and administrators.

• **Account Verification (Optional):** Email or OTP verification can be implemented for added security.

Functionality Flow:

1. User fills the registration form.
2. Input data is validated and stored with encrypted password.
3. On login, credentials are verified and a session is created.
4. Access to features is granted based on user role.

3.2 MOVIE MANAGEMENT MODULE

This This module is responsible for managing movie-related information in the system. It allows administrators to add, update, delete, and display movie details.

Key Features:

• **Add Movie:** Admin can add new movies with details such as title, genre, language, duration.

• **Update & Delete:** Admin can modify or remove movie records.

• **Display Movies:** Movies are displayed for users to browse and select.

• **Movie Details:** Users can view complete information about each movie.

Functionality Flow:

1. Admin adds or updates movie details.
2. Movie data is stored in the database.
3. Users can view available movies on the interface.
4. Selected movie details are shown for booking.

3.3 BOOKING MODULE

The Booking Module is the core component of the system that allows users to book movie tickets by selecting shows and seats.

Key Features:

• **Show Selection:** Users can choose movie show timings and theaters.

• **Seat Selection:** Displays available seats and allows users to select preferred seats.

• **Ticket Booking:** Confirms booking and stores transaction details.

• **Booking Management:** Users can view or cancel their bookings.

Functionality Flow:

1. User selects a movie and show timing.
2. Available seats are displayed.
3. User selects seats and confirms booking.

3.4 ADMIN CONTROL MODULE

This This module provides administrative control over the system, enabling efficient management of movies, users, and bookings.

Key Features:

• **User Management:** Admin can view and manage user accounts.

• **Movie & Show Management:** Add, update, or delete movies and show schedules.

• **Booking Monitoring:** View all booking transactions.

• **System Control:** Ensure smooth functioning and prevent misuse.

Functionality Flow:

1. Admin logs into the system.
2. Accesses admin dashboard.
3. Manages movies, users, and bookings.
4. Performs actions such as updating or deleting records.

3.5 SEARCH & FILTER MODULE

This module enhances user experience by allowing users to search and filter movies based on various criteria.

Key Features:

• **Search Functionality:** Users can search movies by name or keyword.

• **Filter Options:** Movies can be filtered by genre, language, or timing.

• **Dynamic Results:** Displays results instantly without reloading the page.

Functionality Flow:

1. User enters search keyword or selects filters.
2. System processes the query.
3. Matching results are retrieved from the database.
4. Results are displayed dynamically.

3.6 PAYMENT & NOTIFICATION MODULE

This module handles payment processing and user notifications related to bookings.

Key Features:

• **Payment Processing:** Supports different payment methods such as UPI, cards, or wallets.

• **Payment Status Tracking:** Maintains records of successful or failed transactions.

• **Booking Confirmation:** Sends confirmation messages after successful booking.

• **Notifications:** Alerts users about bookings, cancellations, or updates.

Functionality Flow:

1. User proceeds to payment after seat selection.
2. Payment is processed and verified.
3. Booking is confirmed and stored.
4. Notification is sent to the user via email or app interface.

Each module in the Movie Ticket Booking System is independently responsible for a specific functionality that contributes to the overall system. These modules are interconnected and work together as a complete system to provide a smooth, efficient, and secure ticket booking experience. The modular design allows for easy maintenance and future enhancements such as online payment integration, real-time seat availability updates, mobile application support, and advanced features like recommendations and analytics.

CHAPTER 4

DATABASE IMPLEMENTATION

A robust and well-structured database is the backbone of the Movie Ticket Booking System. The database plays a crucial role in storing, organizing, and retrieving data such as user details, movie information, theater data, show schedules, bookings, and payments. This chapter explains how the database is designed and implemented, including table structures, relationships, constraints, and backup and recovery strategies.

4.1 TABLE CREATION SCRIPTS

- The system uses a Relational Database Management System (RDBMS) such as MySQL or PostgreSQL to store structured data. The following tables form the core of the database:

1. Users Table

```
CREATE TABLE Users (
    User_ID INT AUTO_INCREMENT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Email VARCHAR(100) UNIQUE NOT NULL,
    Password VARCHAR(255) NOT NULL,
    Role VARCHAR(20) DEFAULT 'customer'
););
```

- **Purpose:** Stores information about all users including customers and administrators.

2. Movies Table

```
CREATE TABLE Movies (
    Movie_ID INT AUTO_INCREMENT PRIMARY KEY,
    Movie_Name VARCHAR(100) NOT NULL,
    Language VARCHAR(50),
    Genre VARCHAR(50),
    Duration INT
);
```

- **Purpose:** Stores details about movies such as title, genre, language, and duration.

3. Theatres Table

```
CREATE TABLE Theatres (
    Theatre_ID INT AUTO_INCREMENT PRIMARY KEY,
    Theatre_Name VARCHAR(100) NOT NULL,
    Location VARCHAR(100)
);
```

- **Purpose:** Stores information about theaters including name and location.

4. Shows Table

```
CREATE TABLE Shows (
    Show_ID INT AUTO_INCREMENT PRIMARY KEY,
    Movie_ID INT,
    Theatre_ID INT,
    Show_Time DATETIME,
    FOREIGN KEY (Movie_ID) REFERENCES Movies(Movie_ID) ON
DELETE CASCADE,
    FOREIGN KEY (Theatre_ID) REFERENCES
Theatres(Theatre_ID) ON DELETE CASCADE
);
```

- **Purpose:** Stores show timings and links movies with theaters.

5. Bookings Table

```
CREATE TABLE Seats (
    Seat_ID INT AUTO_INCREMENT PRIMARY KEY,
    Show_ID INT,
    Seat_Number VARCHAR(10),
    Seat_Type VARCHAR(20),
    Availability BOOLEAN DEFAULT TRUE,
    FOREIGN KEY (Show_ID) REFERENCES Shows(Show_ID) ON
DELETE CASCADE
);
```

- **Purpose:** Stores booking details including user ID, show ID, seat numbers, and booking date.

6. Seates Table

```
CREATE TABLE Bookings (
    Booking_ID INT AUTO_INCREMENT PRIMARY KEY,
    User_ID INT,
    Show_ID INT,
    Seat_ID INT,
    Booking_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (User_ID) REFERENCES Users(User_ID) ON
DELETE CASCADE,
    FOREIGN KEY (Show_ID) REFERENCES Shows(Show_ID) ON
DELETE CASCADE,
    FOREIGN KEY (Seat_ID) REFERENCES Seats(Seat_ID) ON
DELETE CASCADE
);
```

- **Purpose:** Manages seat information and availability for each show.

7. Payments Table

```
CREATE TABLE Payments (
    Payment_ID INT AUTO_INCREMENT PRIMARY KEY,
    Booking_ID INT,
    Amount DECIMAL(10,2),
    Payment_Method VARCHAR(50),
    Payment_Status VARCHAR(20),
    FOREIGN KEY (Booking_ID) REFERENCES Bookings(Booking_ID)
ON DELETE CASCADE
);
```

- **Purpose:** Stores payment details such as payment method, amount, and status.

4.2 CONSTRAINTS AND RELATIONSHIPS

Implementing primary keys, foreign keys, and constraints ensures that the database maintains data integrity, avoids redundancy, and enforces proper relationships between tables.

Primary Keys

• Each table contains a primary key to uniquely identify records (e.g., User_ID, Movie_ID, Booking_ID, Show_ID).

Foreign Keys

• Foreign keys establish relationships between tables. For example, User_ID in the Bookings table links to the Users table, and Show_ID connects bookings with shows.

Constraints

• NOT NULL constraints ensure essential fields such as user details, movie information, and booking data are always provided.

• UNIQUE constraints prevent duplicate entries, such as duplicate email IDs for users.

• ON DELETE CASCADE ensures that when a record (like a user or movie) is deleted, related records (such as bookings) are automatically removed.

4.3 BACKUP & RECOVERY CONSIDERATIONS

Ensuring data safety and reliability is essential for the Movie Ticket Booking System. Backup and recovery strategies are implemented to prevent data loss due to system failures or errors.

Backup Strategies:

• **Automated Backups:** Regular database backups are scheduled using tools like mysqldump to ensure data is not lost.

• **Cloud Storage (Optional):** Backup files can be stored securely in cloud platforms such as Google Drive or AWS S3.

• **Data Export:** Admins can export booking and user data in formats like CSV or JSON for offline storage.

Recovery Strategies:

• **Point-in-Time Recovery:** The database can be restored to a specific state using logs in case of failure.

• **Transaction Rollback:** In case of failed booking or payment, transactions are rolled back to maintain consistency.

• **Data Restoration:** Backup files can be used to restore lost or corrupted data quickly.

The database implementation in the Movie Ticket Booking System follows best practices of database design, normalization, and relational modeling. It ensures secure storage of user data, efficient handling of movie and booking information, and smooth transaction processing. By enforcing strong constraints, maintaining proper relationships, and implementing backup strategies, the system provides a reliable, scalable, and secure foundation for real-world ticket booking applications.

CHAPTER 5

RESULTS AND DISCUSSION

After the successful development and implementation of the Movie Ticket Booking System, the application was thoroughly tested to evaluate its functionality, performance, and overall user experience. This chapter presents the results obtained from the evaluation and discusses how effectively the system meets its intended objectives, along with identifying possible areas for further enhancement and improvement.

5.1 Performance Analysis

The performance of the Movie Ticket Booking System was analyzed based on several key parameters, including response time, booking operations, database efficiency, user interface responsiveness, and system reliability. The frontend performance of the system was found to be highly efficient, as the user interface was responsive across different devices such as desktops, tablets, and mobile phones. The page load time was minimal, typically within 2 to 3 seconds, which was achieved through optimized design and efficient use of CSS and JavaScript. Navigation between various pages, including movie selection, seat booking, and confirmation, was smooth and user-friendly. Additionally, the seat selection interface responded quickly, offering users a real-time booking experience.

From a backend perspective, the system demonstrated strong processing capabilities. Core operations such as booking tickets, canceling bookings, and retrieving movie details were executed accurately without errors. User authentication processes, including login and registration, functioned securely with proper password encryption and session management. The backend efficiently handled multiple requests simultaneously with minimal delay. Furthermore, booking validation mechanisms ensured that issues such as seat duplication or double booking did not occur.

Database operations were also highly efficient due to proper database design. SQL queries used for retrieving movies, shows, and booking details were executed quickly and effectively. Insert and update operations related to bookings and payments were consistent and reliable. The use of primary and foreign keys ensured strong relationships between database tables, while data integrity was maintained through the implementation of constraints and cascading rules.

In terms of security, the system incorporated essential measures to protect user data. Passwords were securely stored using hashing techniques, and input validation was implemented to prevent invalid or malicious data entry. Additionally, the use of prepared statements helped protect the system against SQL injection attacks. Regarding scalability, initial observations indicated that the system could handle multiple users and booking operations efficiently during testing. It supported several movies, shows, and concurrent bookings without noticeable performance degradation. However, the system can be further enhanced by implementing advanced techniques such as caching, indexing, and load balancing to support large-scale usage.

5.2 Observations and Interpretations

The evaluation of the system provided valuable insights into user experience, administrative functionality, and overall system stability. From a user experience perspective, the system received positive feedback. Users found the application easy to use and navigate, and the process of browsing movies and booking tickets was considered simple and convenient. The real-time seat selection feature significantly improved user satisfaction by providing immediate feedback during booking.

However, some challenges were observed during testing. A few users indicated that clearer seat availability indicators would improve usability, suggesting that minor user interface enhancements could make the system more intuitive. Additionally, the integration of online payment features was identified as an important improvement to enhance real-world applicability and convenience.

From the administrative perspective, the system proved to be effective in managing movies, shows, and bookings. Administrators were able to perform their tasks efficiently using the admin dashboard, which provided centralized control over system operations. Nevertheless, the inclusion of advanced features such as analytics, including insights on the most booked movies and peak booking times, could further enhance decision-making capabilities.

In terms of functional stability, all major modules of the system, including user management, movie management, booking system, and admin controls, functioned correctly without any errors. The system handled booking operations reliably without crashes or inconsistencies, and proper error handling mechanisms ensured smooth execution of all processes. Visually, the system featured a clean, simple, and easy-to-understand user interface. The responsive design ensured that the application remained accessible and user-friendly across a wide range of devices, contributing to a positive overall user experience.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 SUMMARY OF OUTCOMES

The system’s performance was evaluated based on three main criteria: booking accuracy, real-time responsiveness, and usability from a user-experience perspective on the web interface. The Movie Ticket Booking System successfully provided an efficient platform for users to browse movies, select show timings, choose seats, and complete bookings with ease. The booking module ensured accurate seat allocation and prevented duplicate bookings through proper validation mechanisms. The database operations, including storing and retrieving user data, movie details, and booking records, were performed efficiently using structured queries and relational design. The web interface delivered a smooth and responsive experience, with most operations such as movie selection and booking confirmation completed within a few seconds. Overall, the system achieved a balanced combination of performance, reliability, and ease of use, making it practical for real-time applications. However, certain limitations were identified, such as the lack of integrated online payment systems and limited scalability under heavy user traffic. Additionally, advanced features like personalized recommendations and detailed analytics were not included in the current version.

6.2 FUTURE SCOPE AND ENHANCEMENTS

The Movie Ticket Booking System has significant potential for further enhancement and expansion to meet real-world demands. Future improvements can include the integration of online payment systems such as UPI, debit/credit cards, and digital wallets to enable seamless transactions. The system can be enhanced with real-time notifications through email or SMS for booking confirmations and updates. Advanced features such as movie recommendations based on user preferences, search and filtering options, and dynamic seat selection interfaces can improve user experience. The addition of an analytics dashboard for administrators can help track booking trends, popular movies, and user activity. Furthermore, the system can be extended into a mobile application using frameworks like React Native or Flutter to increase accessibility. Performance can be improved by implementing caching, indexing, and load balancing techniques to handle large-scale users. The modular design of the current system allows these enhancements to be integrated with minimal changes, ensuring scalability and adaptability. In conclusion, the project provides a strong foundation for a real-world movie ticket booking application while offering ample opportunities for future growth and technological advancemen.

APPENDICES

APPENDIX A

SOURCE CODE

This section presents key source code snippets from the Blogging Platform application. The code is written using PHP for the server-side scripting and MySQL for the backend database.. The source code follows a modular and secure approach, with appropriate validation and database interaction.

A.1 User Registration Module

Purpose: Allows new users to register and securely stores their credentials in the database**. Php :**

```
<?php
$conn = mysqli_connect("localhost", "root", "", "movie_db");

if(isset($_POST['register'])) {
   $name = $_POST['name'];
   $email = $_POST['email'];
   $password = password_hash($_POST['password'], PASSWORD_DEFAULT);

   $sql = "INSERT INTO Users (Name, Email, Password) VALUES ('$name',
'$email', '$password')";
   mysqli_query($conn, $sql);
}
?>
```

Explanation:

- Accepts name, email, and password from the form.
- Password is hashed using password_hash() to ensure security.
- Data is stored in the Users table.

A.2 User Login Module

Purpose: Authenticates users and starts a session.

```
<?php
session_start();
$conn = mysqli_connect("localhost", "root", "", "movie_db");

if(isset($_POST['login'])) {
   $email = $_POST['email'];
   $password = $_POST['password'];

   $result = mysqli_query($conn, "SELECT * FROM Users WHERE
Email='$email'");
   $user = mysqli_fetch_assoc($result);

   if($user && password_verify($password, $user['Password'])) {
      $_SESSION['user_id'] = $user['User_ID'];
      header("Location: home.php");
   } else {
      echo "Invalid login!";
   }
}
```

Explanation:

- Retrieves user details using email.
- Verifies password using password_verify().
- Creates session for logged-in user.

A.3 Movie Add Module

Purpose: Allows admin to add new movies.

```
<?php
$conn = mysqli_connect("localhost", "root", "", "movie_db");

if(isset($_POST['add_movie'])) {
    $name = $_POST['movie_name'];
    $genre = $_POST['genre'];
    $duration = $_POST['duration'];

    $sql = "INSERT INTO Movies (Movie_Name, Genre, Duration)
            VALUES ('$name', '$genre', '$duration')";
    mysqli_query($conn, $sql);
}
?>
```

Explanation:

- Admin enters movie details.
- Links Data is inserted into Movies table.

A.4 Display Movie Module

Purpose: Fetches and displays available movies.

```
<?php
$conn = mysqli_connect("localhost", "root", "", "movie_db");

$result = mysqli_query($conn, "SELECT * FROM Movies");

while($row = mysqli_fetch_assoc($result)) {
   echo "<h3>".$row['Movie_Name']."</h3>";
   echo "<p>Genre: ".$row['Genre']."</p>";
}
```

Explanation:

- Retrieves all movies from database.
- Displays movie details to users.

A.5 Ticket Booking Module

Purpose: Allows users to book movie tickets.

```
<?php
session_start();
$conn = mysqli_connect("localhost", "root", "", "movie_db");

if(isset($_POST['book'])) {
   $user_id = $_SESSION['user_id'];
   $show_id = $_POST['show_id'];
   $seat_id = $_POST['seat_id'];

   $sql = "INSERT INTO Bookings (User_ID, Show_ID, Seat_ID)
          VALUES ('$user_id', '$show_id', '$seat_id')";
   mysqli_query($conn, $sql);
}
?>
```

Explanation:

- Takes selected show and seat details.
- Links booking with user ID.
- Stores booking in Bookings table.

A.6 Payment Module

Purpose: Handles payment for bookings.

```
<?php
$conn = mysqli_connect("localhost", "root", "",
"movie_db");

if(isset($_POST['pay'])) {
   $booking_id = $_POST['booking_id'];
   $amount = $_POST['amount'];
   $method = $_POST['method'];

   $sql = "INSERT INTO Payments (Booking_ID, Amount,
Payment_Method, Payment_Status)
         VALUES ('$booking_id', '$amount', '$method',
'Success')";
   mysqli_query($conn, $sql);
}
?>
```

Explanation:

- Stores payment details for each booking.
- Marks payment status as successful

.

APPENDIX B – Screenshots

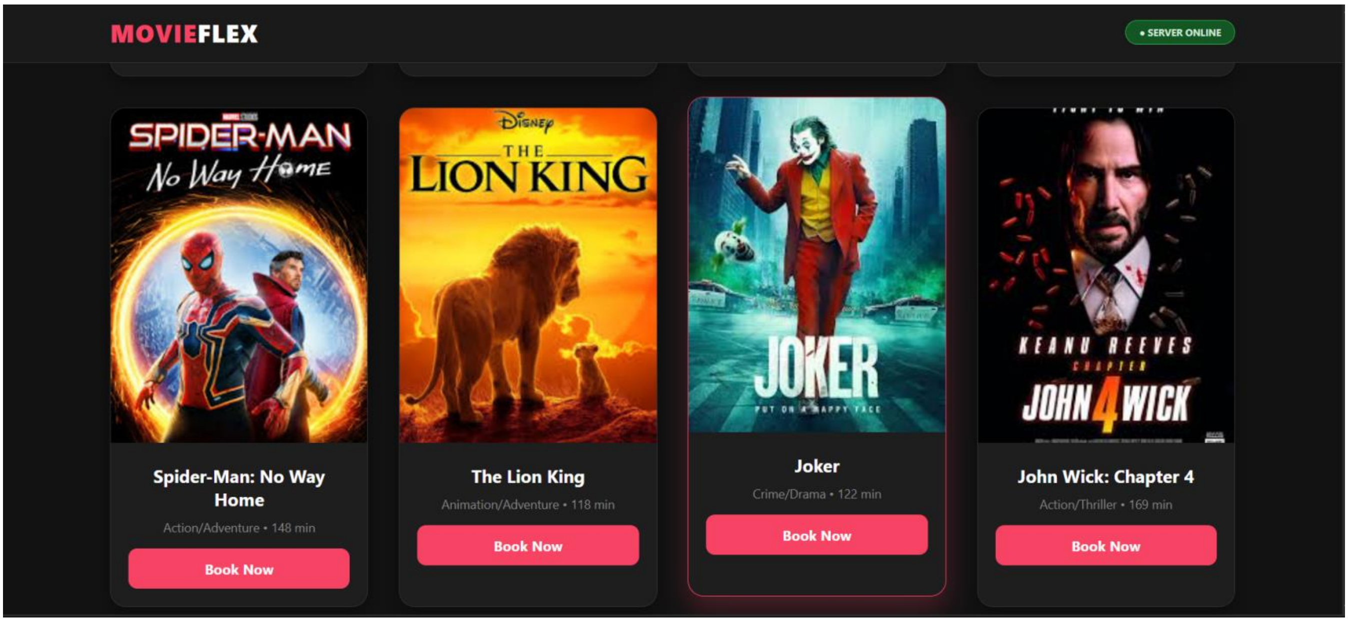

Figure B.1 LOGIN PAGE

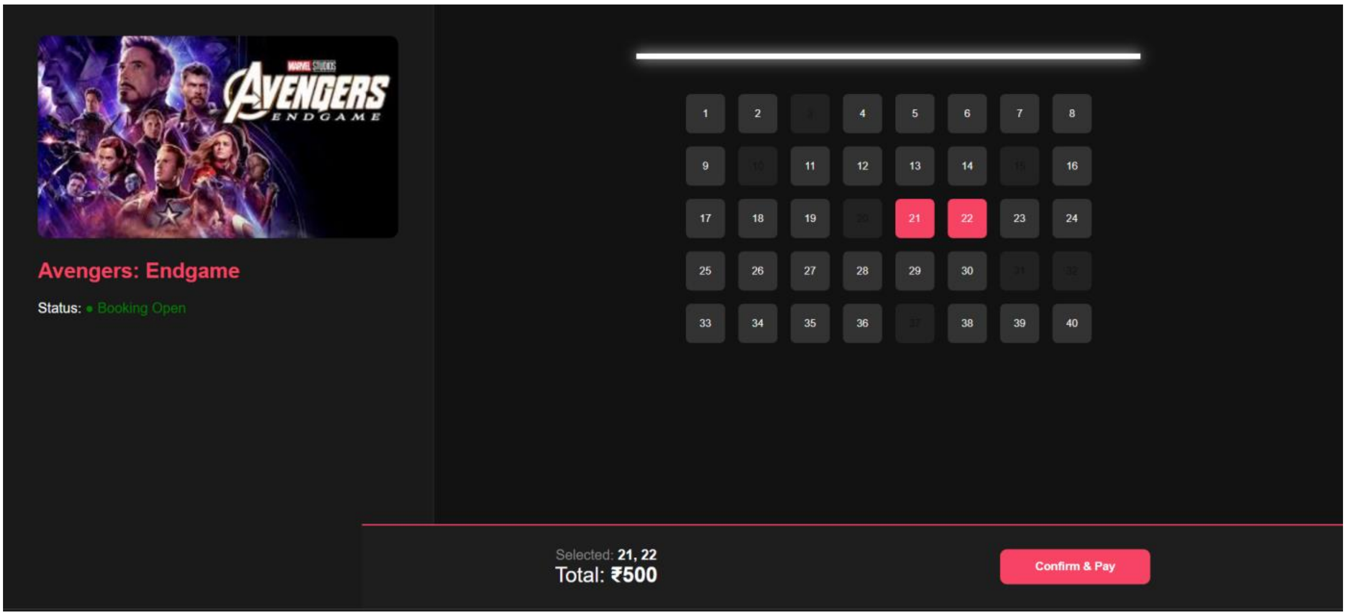

Figure B.2 USER INTERFACE PAGE

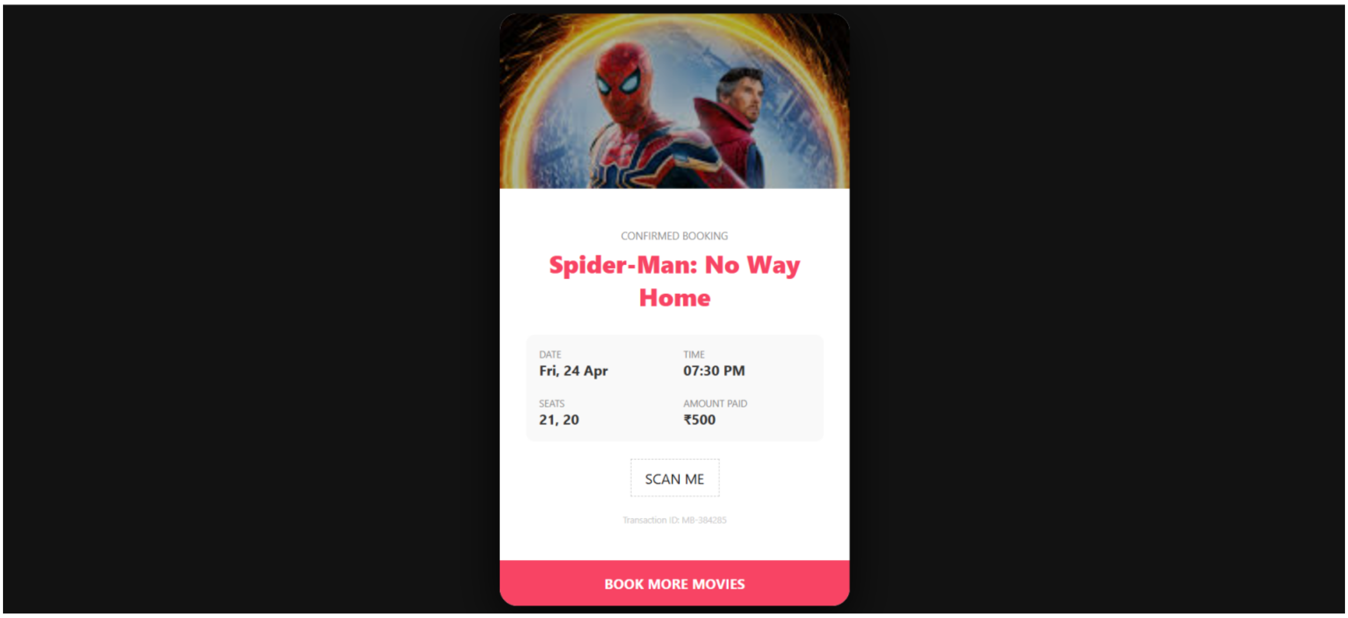

Figure B.3 TICKET CONFIRMATION PAGE

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